

1、创建（初始化）模型

题目中要求使用的模型类名

```
rf_model = RandomForestRegressor(n_estimators=100, random_state=42)
```

2、训练模型

.fit(X_train, y_train) # X_train: 特征变量的训练集, y_train: 目标变量的训练集

```
rf_model.fit(X_train, y_train)
```

3、模型预测（测试）

.predict(X_test) # X_test: 特征变量的测试集; y_pred: 是模型预测的结果

```
y_pred = rf_model.predict(X_test)
```

4、测试分析

训练集得分: 模型变量名.score(X_train, y_train)

测试集得分: 模型变量名.score(X_test, y_test)

均方误差: mean_squared_error(y_test, y_pred)

决定系数: r2_score(y_test, y_pred)

```
train_score = rf_model.score(X_train, y_train) #训练集分数
test_score = rf_model.score(X_test, y_test) #测试集分数
mse = mean_squared_error(y_test, y_pred) #均方误差
r2 = r2_score(y_test, y_pred) #决定系数
```

5、保存模型

pickle.dump(模型的变量名, 保存到文件名)

```
with open('2.2.3_model.pkl', 'wb') as model_file:
    pickle.dump(rf_model, model_file)
```